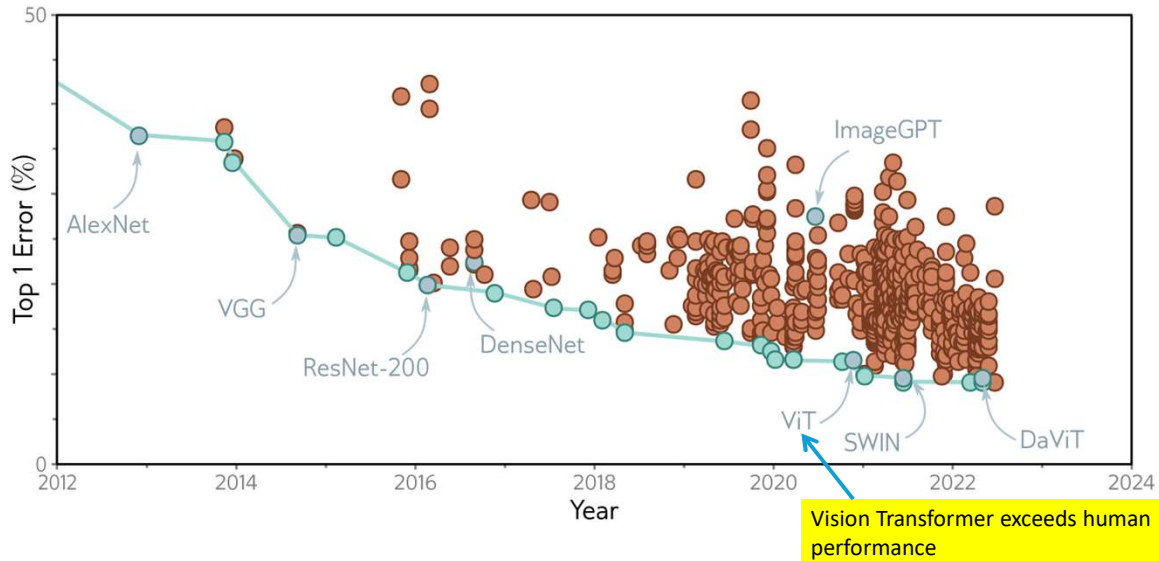


Transformers (Continued)

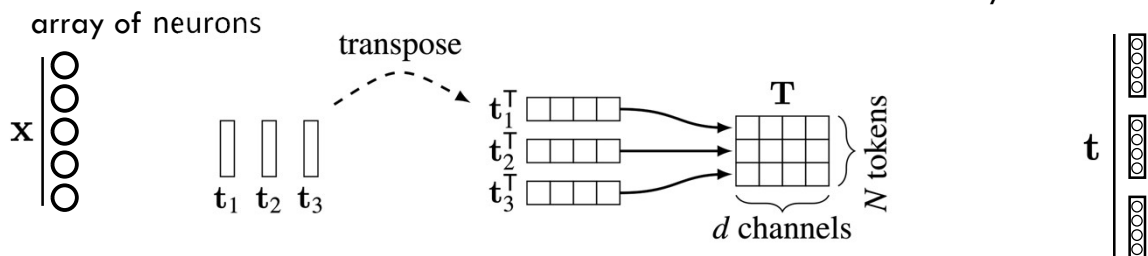
- Three concepts
1. Tokens
 2. Attention
 3. Positional Encoding



1

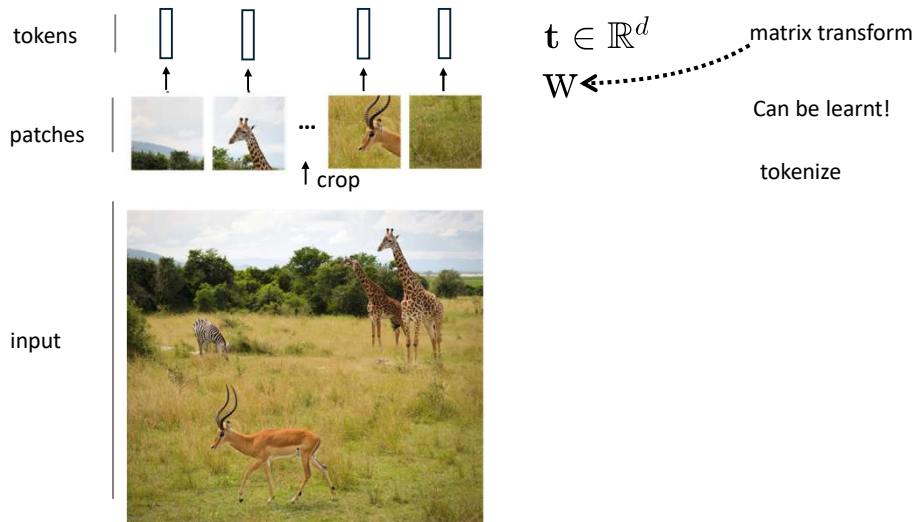
A New Data Type: Tokens

- A token can be thought of as a vector of neurons ("patch")
 - But the connotation is that a token is an encapsulated bundle of information; with transformers we will operate over tokens rather than over neurons.
- We can also have an array of tokens
- CNN interpretation: Disguised channels



2

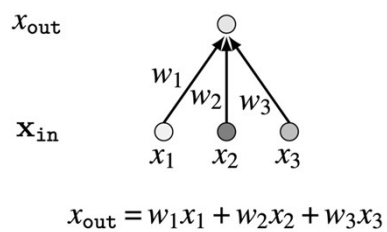
How to get tokens?



3

Operation on Tokens

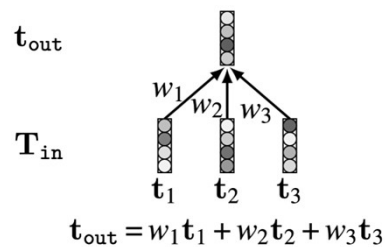
Linear combination of neurons



$$x_{\text{out}}[i] = \sum_{j=1}^N w_{ij} x_{\text{in}}[j]$$

$$\mathbf{x}_{\text{out}} = \mathbf{W} \mathbf{x}_{\text{in}}$$

Linear combination of tokens



$$\mathbf{T}_{\text{out}}[i, :] = \sum_{j=1}^N w_{ij} \mathbf{T}_{\text{in}}[j, :]$$

$$\mathbf{T}_{\text{out}} = \mathbf{W} \mathbf{T}_{\text{in}}$$

4

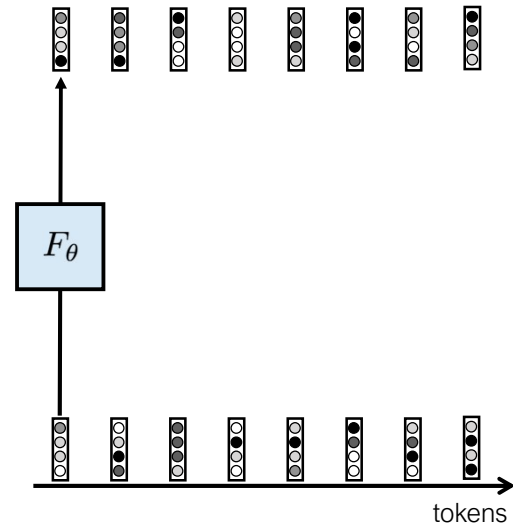
Token-wise nonlinearity

$$\mathbf{x}_{\text{out}} = \begin{bmatrix} \text{relu}(x_{\text{in}}[0]) \\ \vdots \\ \text{relu}(x_{\text{in}}[N-1]) \end{bmatrix}$$

F is typically an MLP

$$\mathbf{T}_{\text{out}} = \begin{bmatrix} F_{\theta}(\mathbf{T}_{\text{in}}[0, :]) \\ \vdots \\ F_{\theta}(\mathbf{T}_{\text{in}}[N-1, :]) \end{bmatrix}$$

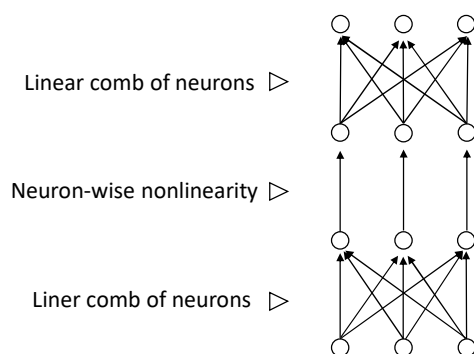
The abstractions of the data you want from an ANN is buried in this MLP



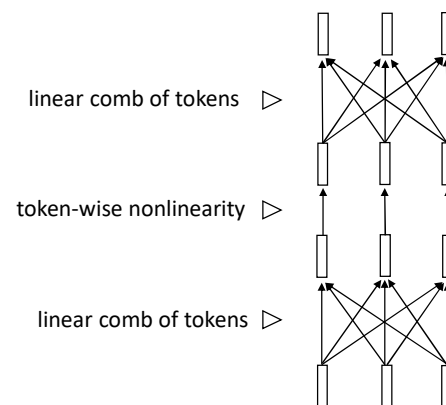
5

Token nets

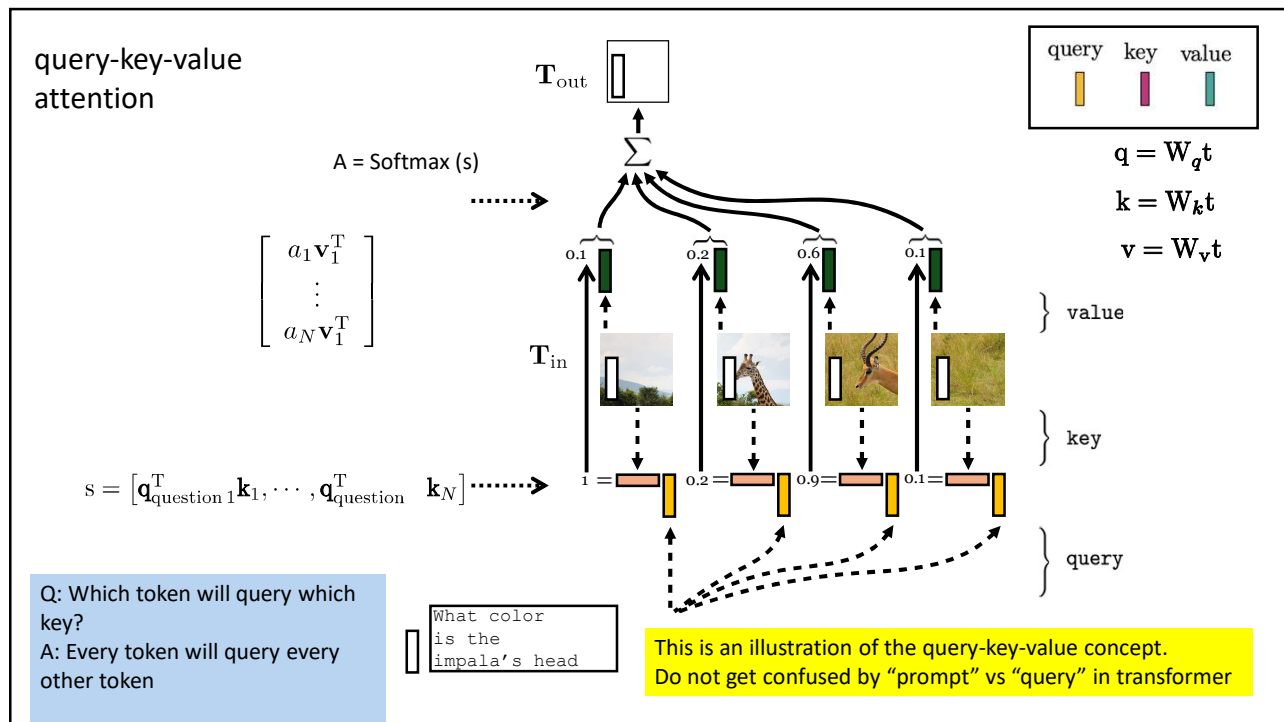
Neural Net



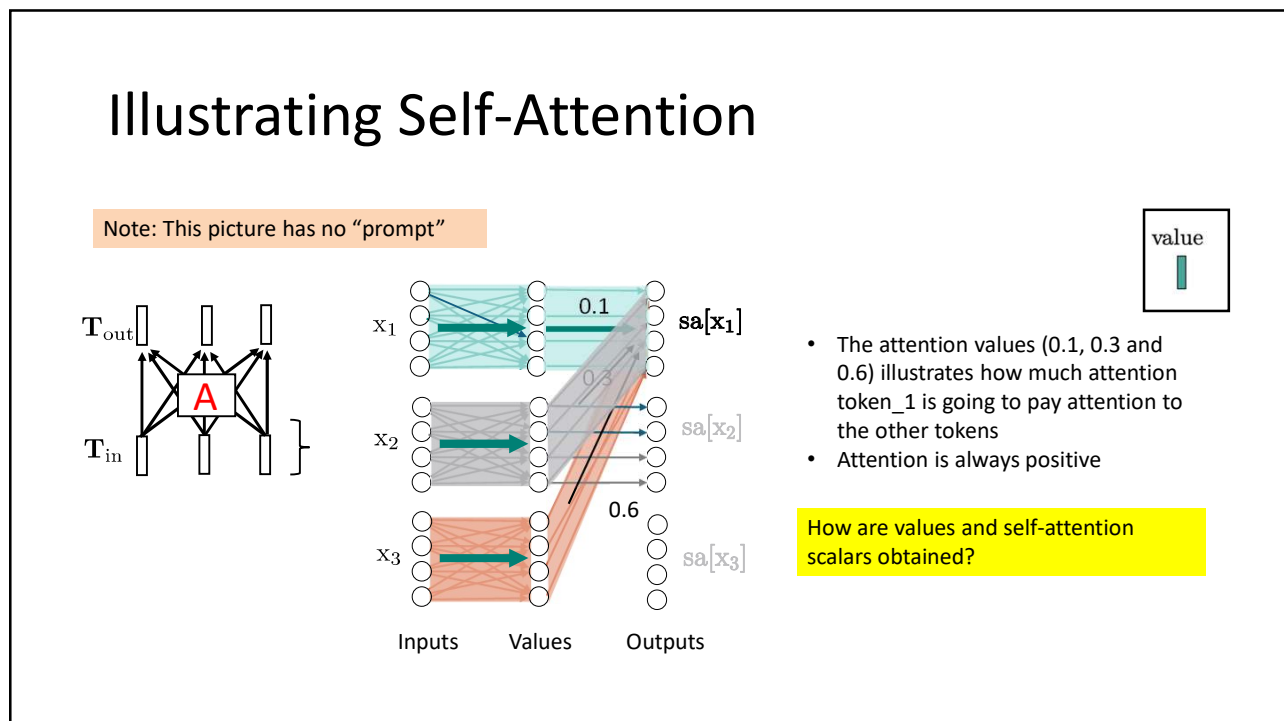
Token Net



6

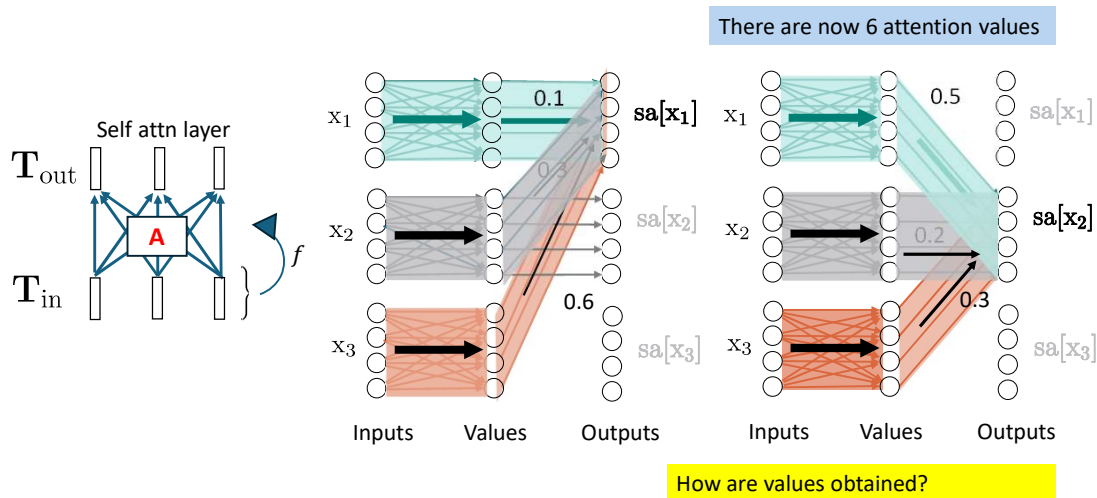


8



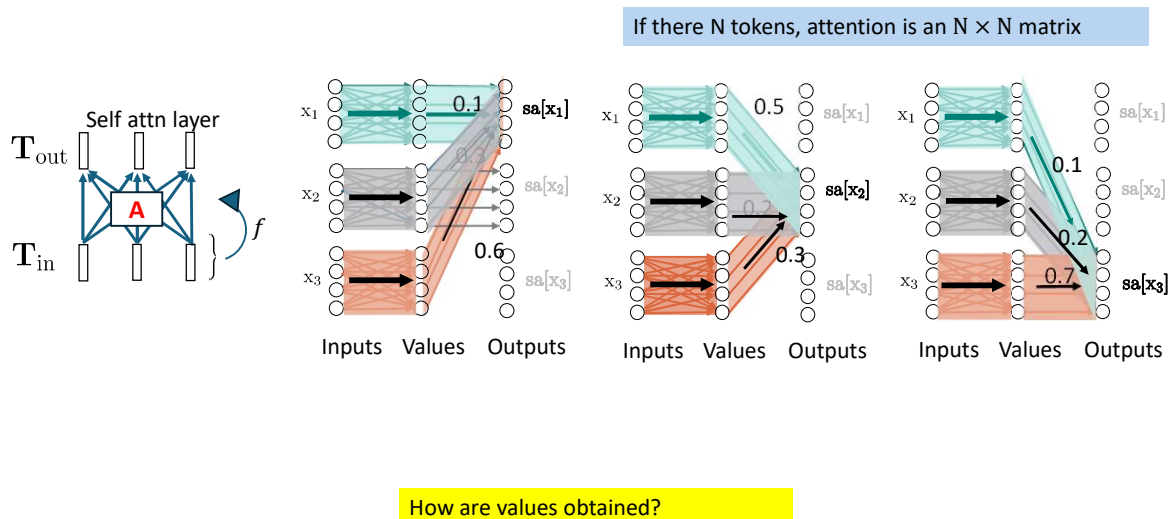
9

Illustrating Self-Attention



10

Illustrating Self-Attention

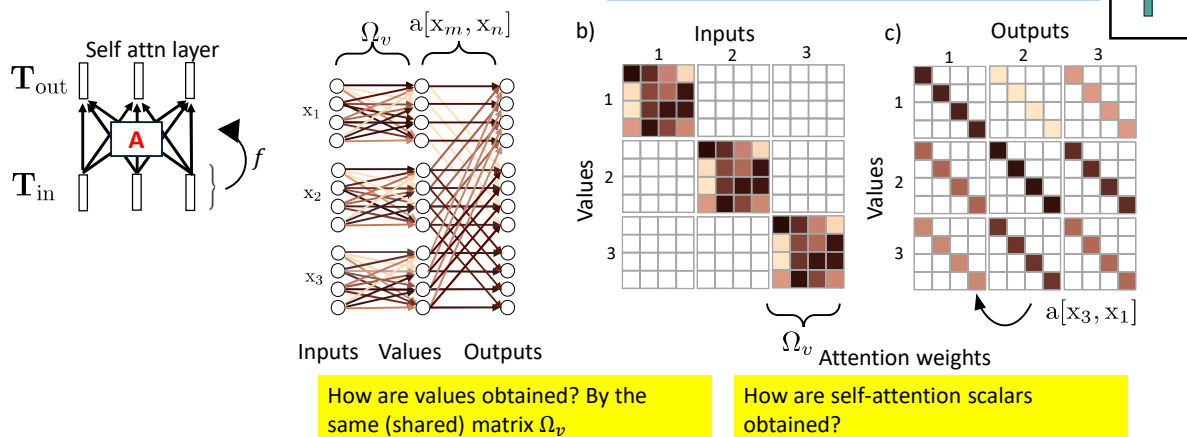


11

Values and Attention

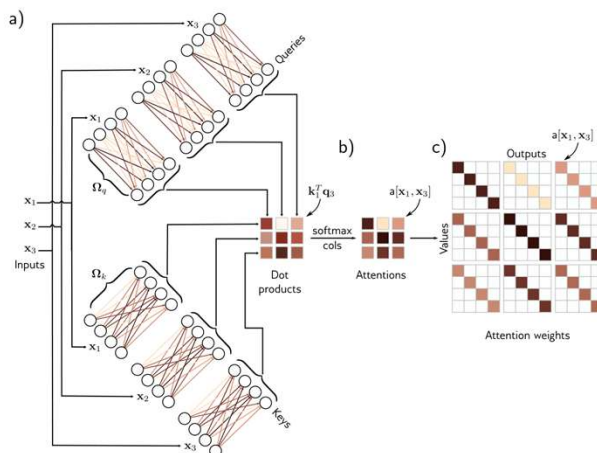
The value matrix is shared across all tokens, and is independent of N

The attention matrix does not depend on the dimension. For 12 neurons and $N = 3$ matrix, there are 9 distinct self-attention scalar



12

Self-Attention Matrix

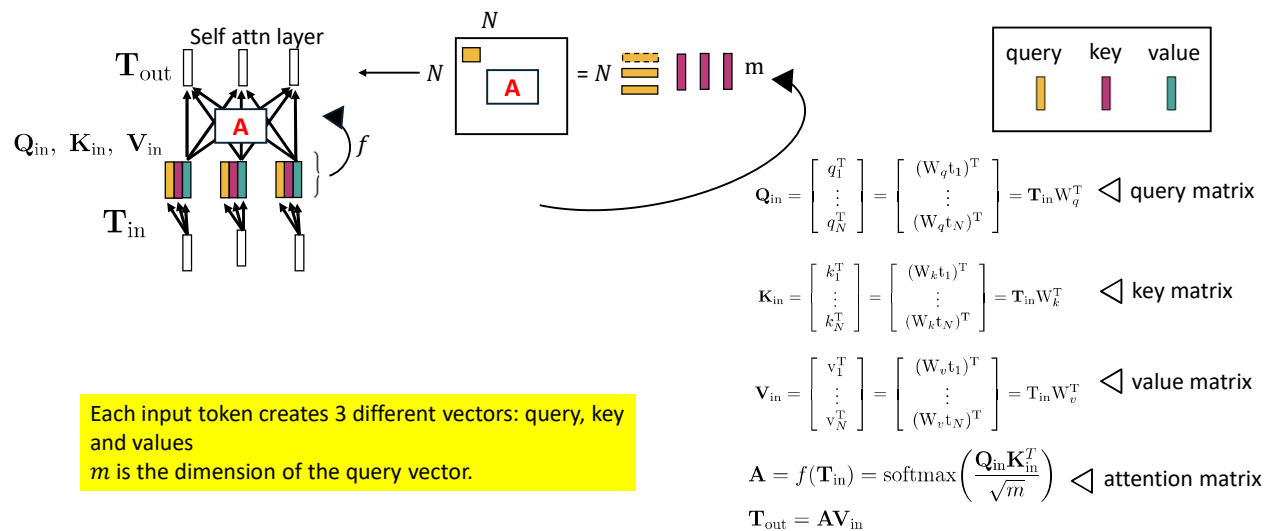


From left to right

- 3 tokens
- (Top Left): Queries, Bottom (Keys)
- (b, Middle): dot product, and softmax to get attention
- (c, Right) Attention scalars are applied individually on each item in a token

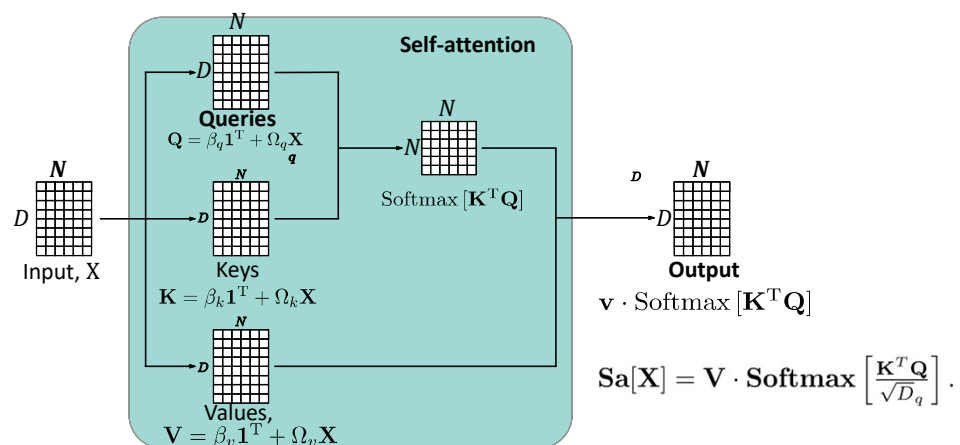
13

Self-Attention layer (expanded)



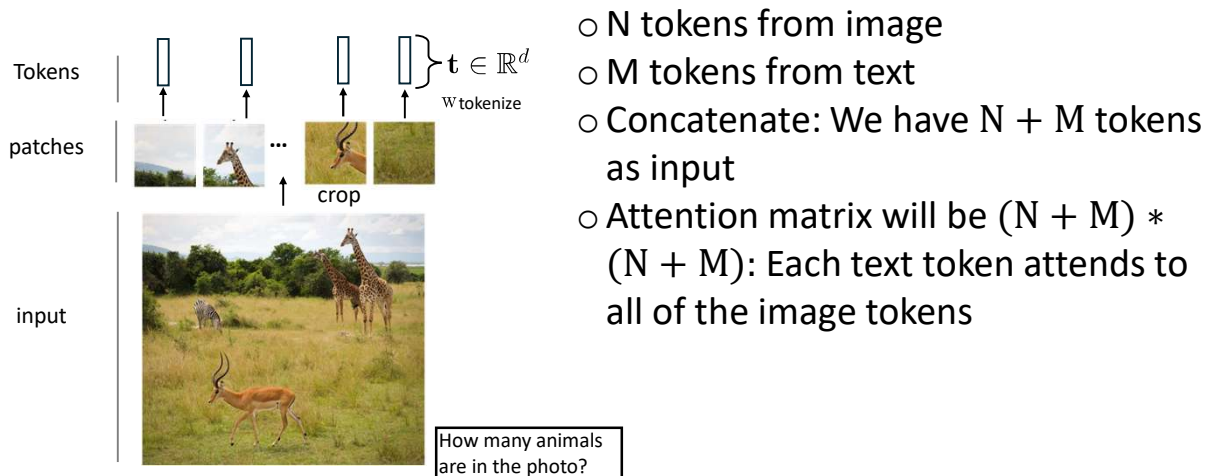
14

Matrix form



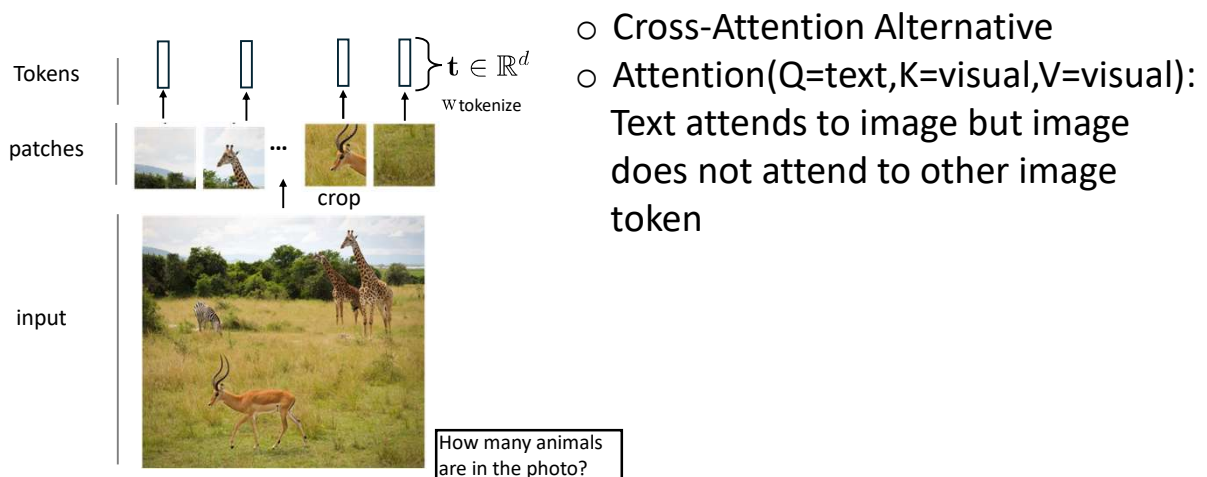
15

VQA: Visual Question Answer



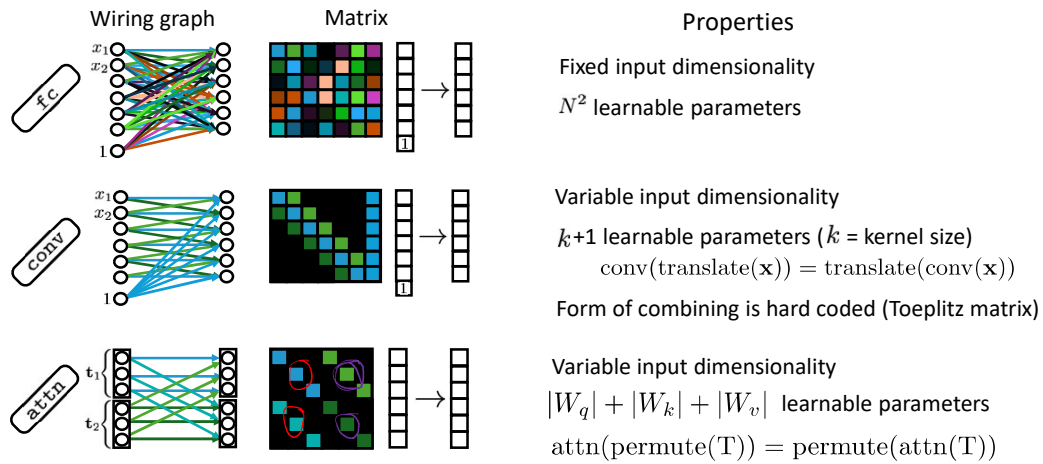
16

VQA: Visual Question Answer



17

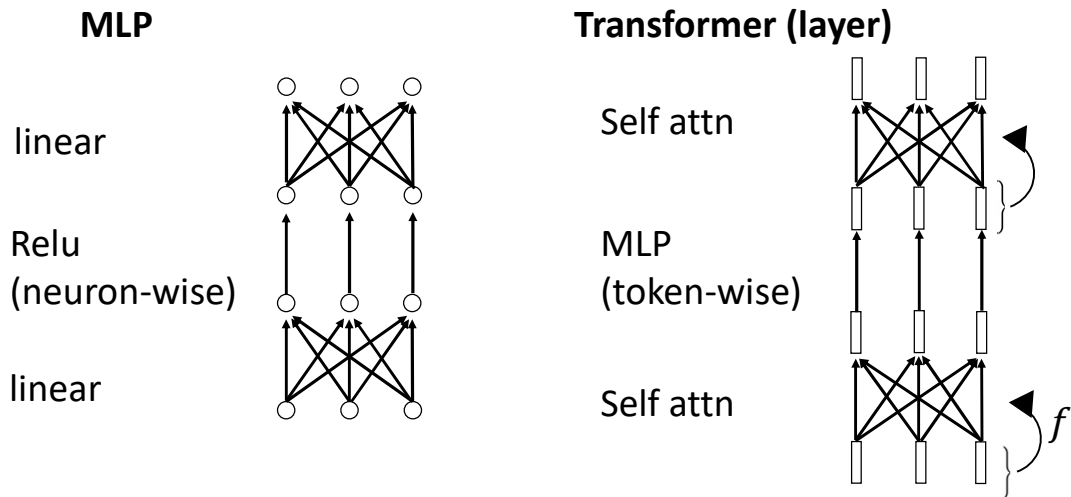
A family of linear layers



18

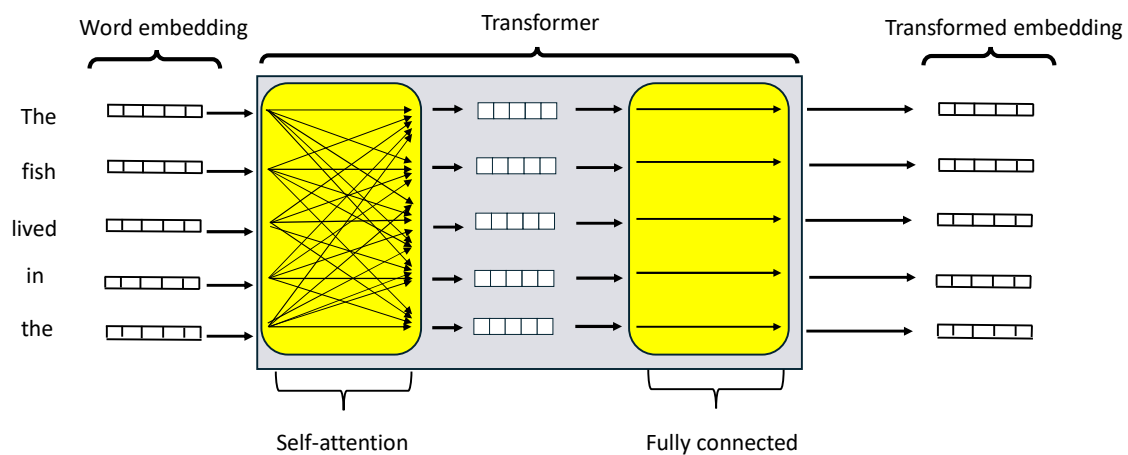
19

Transformer Layer



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Example: Vanilla transformer layer



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Multihead self-attention (MSA)

- Rather than having just one way of attending, why not have k ?
 - Each gets its own parameterized query(), key(), value() functions.
 - (weighted) sum for the output token code
- Run them all in parallel

v is the dimension
of value vector

$$\mathbf{T}_{\text{out}}^i = \text{attn}^i(\mathbf{T}_{\text{in}}) \quad \text{for } i \in \{1, \dots, k\}$$

$$\mathbf{T}_{\text{out}} = \begin{bmatrix} \mathbf{T}_{\text{out}}^1[0, :] & \dots & \mathbf{T}_{\text{out}}^k[0, :] \\ \vdots & & \vdots \\ \mathbf{T}_{\text{out}}^1[N-1, :] & \dots & \mathbf{T}_{\text{out}}^k[N-1, :] \end{bmatrix}$$

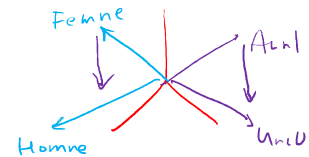
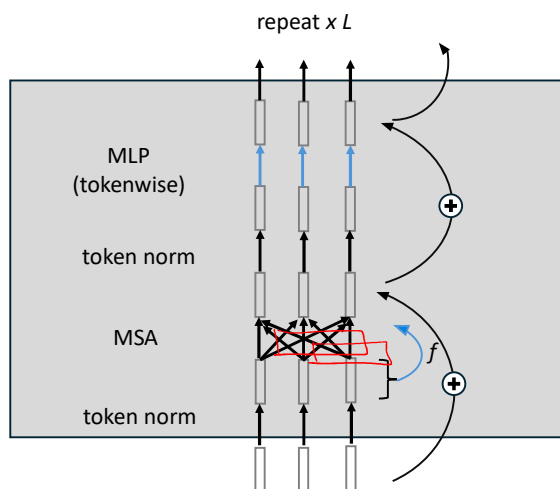
$$\triangleleft \mathbf{T}_{\text{out}} \in \mathbf{R}^{N \times kv}$$

$$\mathbf{T}_o = \mathbf{T}_{\text{out}} \mathbf{W}_{\text{MSA}}$$

$$\triangleleft \mathbf{W}_{\text{MSA}} \in \mathbf{R}^{kv \times d}$$

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Transformer (ViT)



- token norm is an ANN training technique of normalization like layer normalization and batch normalization
- Blue: learnable parameters

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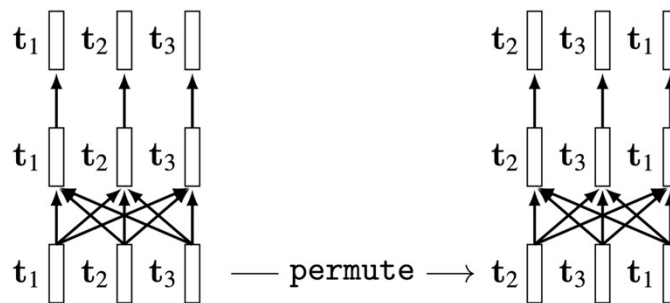
New idea #3: positional encoding

The food was bad, not good at all

The food was good, not bad at all

26

Permutation equivariance



$$F_{\theta}(\text{permute}(\mathbf{T}_{\text{in}})) = \text{permute}(F_{\theta}(\mathbf{T}_{\text{in}}))$$

$$\text{attn}(\text{permute}(\mathbf{T}_{\text{in}})) = \text{permute}(\text{attn}(\mathbf{T}_{\text{in}}))$$

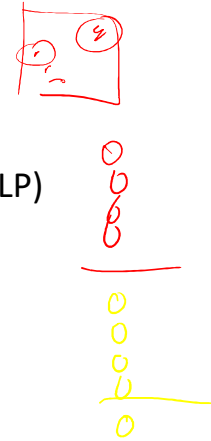
Set2Set

$$\text{transformer}(\text{permute}(\mathbf{T}_{\text{in}})) = \text{permute}(\text{transformer}(\mathbf{T}_{\text{in}}))$$

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How to achieve shift-variant?

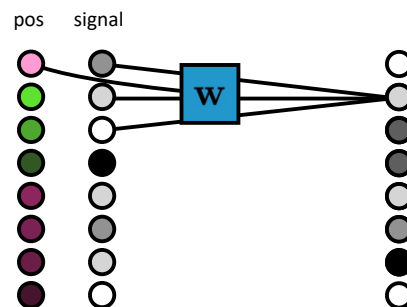
- Can use an architecture that is not shift invariant (e.g., MLP)
- Usual pitfalls of MLP



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Positional Encoding

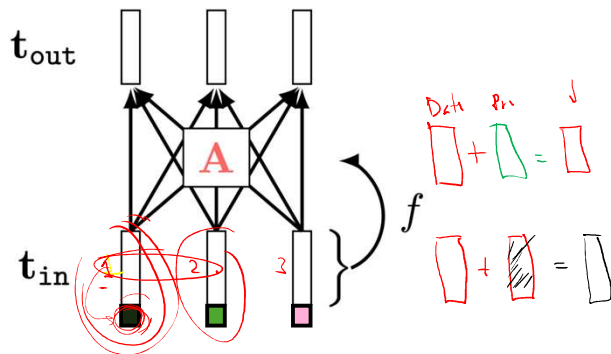
- Can use an architecture that is not shift invariant (e.g., MLP)
- Add location information to the input (CNN example)



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Positional Encoding

- Can use an architecture that is not shift invariant (e.g., MLP)
- Add location information to the input (transformer)



Problem
Input size has
become larger

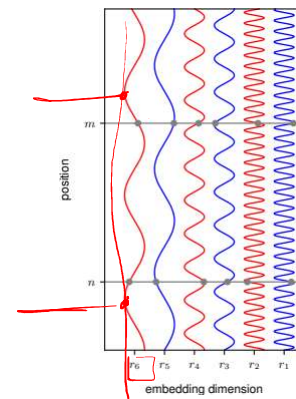
Alternative: For
each position n

$$\tilde{x}_n = x_n + r_n$$

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Positional Encoding Properties

- Unique representation for each position
 - Should be bounded
 - Generalize to longer sequences
 - Consistent way of expressing the number of steps between any two input vectors irrespective of absolute positions
 - Relative positions are important



$$\tilde{x}_n = \widetilde{x_n} + r_n$$

L is a parameter (10000
in the *Attention* paper)

$$r_{ni} = \begin{cases} \sin \left(L^{i/d_{\text{model}}} \right) & \text{if } i \text{ is even} \\ \cos \left(L^{(i-1)/d_{\text{model}}} \right) & \text{if } i \text{ is odd} \end{cases}$$

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Fourier positional codes for images

- Represent coordinates on Fourier basis

